Think — angles in design

Tiles, patterns and the *360°* that has to be filled at every meeting point.

LESSON 88 | 1 × 40 MIN | MATEMATIKA 6, O‘YLAB KO‘R | S7 5.3

LESSON CLOCK

10' | 12' | 12' | 6'

The angles of a regular polygon	10 min
Which shapes tile	12 min
Why pentagons fail	12 min
Ornament	6 min

LEARNING OBJECTIVES

- Find the angles of a regular polygon.
- Decide which regular polygons tile the plane.
- Explain a tiling by the angles at a meeting point.
- Recognise the geometry in Uzbek ornament.

TEXTBOOK

National	pp. 247–249
Cambridge	Stage 7, pp. 62–65
Workbook	Exercise 5.4

01 Explanation

The angles of a regular polygon

$$\text{interior angle} = \frac{180°(n-2)}{n}$$

POLYGON	N	ANGLE SUM	EACH ANGLE
triangle	3	180°	60°
square	4	360°	90°
pentagon	5	540°	108°
hexagon	6	720°	120°
octagon	8	1080°	135°

THE ANGLE SUM COMES FROM TRIANGLES

An n-sided polygon splits into $n-2$ triangles from one vertex, so its angles add to $180°(n-2)$. A regular one shares that equally.

Which shapes tile

Tiles meet without gaps only if the angles at each meeting point add to exactly *360°*.

SHAPE	ANGLE	$360 \div ANGLE$	TILES?
triangle	*60°*	*6*	yes
square	*90°*	*4*	yes
pentagon	*108°*	*3.33…*	no
hexagon	*120°*	*3*	yes
octagon	*135°*	*2.67…*	no, alone

THE TEST IS A WHOLE-NUMBER DIVISION

Only three regular polygons tile the plane by themselves: the triangle, the square and the hexagon. Bees use the third of them.

Why pentagons fail

PENTAGONS AT A POINT	TOTAL	VERDICT
3	*324°*	a gap of *36°*
4	*432°*	an overlap of *72°*

Octagons alone fail too — but octagons and squares together work: $135 + 135 + 90 = 360$, which is the pattern on countless tiled floors.

MIXED TILINGS OPEN THE QUESTION UP AGAIN

Squares and triangles: $90 + 90 + 60 + 60 + 60 = 360$. Hexagons and triangles: $120 + 120 + 60 + 60 = 360$. Each combination is a small piece of arithmetic with a floor pattern as its answer.

Ornament

WHERE	SHAPES USED	ANGLE FACT BEHIND IT
a girih tile panel	stars and hexagons	angles meeting at $360°$
a brick floor	rectangles	$90° \cdot 4$
a honeycomb	hexagons	$120° \cdot 3$
a tiled dome	triangles and squares	mixed tilings

THE ORNAMENT OF SAMARKAND IS ANGLE ARITHMETIC

The star patterns on the madrasahs are built from polygons whose angles fit exactly round a point. The craftsmen worked it out with compasses and a straightedge, five centuries before the formula was written the way it is above.

02 Worked examples

EXAMPLE 1 **Find each interior angle of a regular hexagon.**

1. Angle sum: $180 \cdot (6 - 2) = 720°$.
2. Shared equally between six angles.
3. $720 \div 6 = 120°$

ANSWER

$120°$

EXAMPLE 2 **Do regular pentagons tile the plane?**

1. Each angle is $108°$.
2. $360 \div 108 = 3.33\ldots$ — not a whole number.
3. Three leave a gap of $36°$; four overlap.
4. So no.

ANSWER

No

EXAMPLE 3 **Show that octagons and squares tile together.**

1. An octagon angle is $135°$, a square angle $90°$.
2. $135 + 135 + 90 = 360$
3. Two octagons and a square meet exactly at each point.

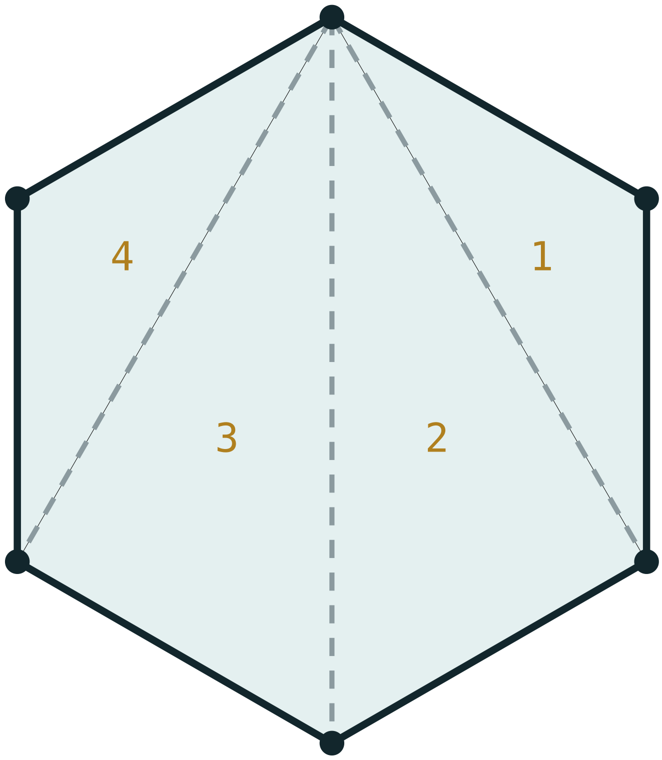
ANSWER

They tile

03 Interactive model

Cut out card pentagons and let the class try to tile a desk; the stubborn gap is a better memory than the arithmetic that predicts it.

Angle sum of a polygon



$$180° \times (6 - 2) = 720°$$

sides n **6** | triangles **4** | interior sum **720°** | each angle **120°** | exterior sum **360°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Regular polygon	Muntazam ko‘pburchak	Правильный многоугольник
Tiling	Qoplama	Замощение
Meeting point	Uchrashuv nuqtasi	Точка стыка
Interior angle	Ichki burchak	Внутренний угол
Hexagon	Olti burchak	Шестиугольник
Pentagon	Besh burchak	Пятиугольник
Ornament	Naqsh	Орнамент
Symmetry	Simmetriya	Симметрия

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The angle sum of a pentagon is:

- *360°*
- *540°*
- *720°*
- *900°*

Each angle of a regular hexagon is:

- *108°*
- *120°*
- *135°*
- *144°*

Tiles meet without gaps when the angles add to:

- *180°*
- *270°*
- *360°*
- *540°*

Regular pentagons:

- tile
- do not tile
- tile with triangles only
- tile in threes

How many regular polygons tile alone?

- 1
- 2
- 3
- 4

Two octagons and a square meet at:

- $300°$
- $330°$
- $360°$
- $390°$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. The angle sum of a quadrilateral

   ANSWER *360°*

2. The angle sum of a pentagon

   ANSWER *540°*

3. The angle sum of a hexagon

   ANSWER *720°*

4. Each angle of an equilateral triangle

   ANSWER *60°*

5. Each angle of a regular pentagon

   ANSWER *108°*

6. Each angle of a regular hexagon

   ANSWER *120°*

7. Each angle of a regular octagon

   ANSWER *135°*

Medium · the standard question

7 PROBLEMS

1. Do triangles tile?

   ANSWER Yes — six at a point

2. Do squares tile?

   ANSWER Yes — four at a point

3. Do pentagons tile?

   ANSWER No

4. Do hexagons tile?

   ANSWER Yes — three at a point

5. Do octagons tile alone?

   ANSWER No

6. Octagons and squares together?

   ANSWER Yes — $135 + 135 + 90$

7. The angle sum of a decagon

   ANSWER $1440°$

Hard · stretch and reason

7 PROBLEMS

1. Each angle of a regular decagon

   ANSWER $144°$

2. Do decagons tile alone?

   ANSWER No — $360 \div 144 = 2.5$

3. Squares and triangles at a point: one arrangement

   ANSWER $90 + 90 + 60 + 60 + 60 = 360$

4. Hexagons and triangles at a point

   ANSWER $120 + 120 + 60 + 60 = 360$

5. A regular polygon with interior angle $150°$: how many sides?

   ANSWER 12

6. Why does the interior angle grow with n?

   ANSWER The angle sum grows faster than the number of angles

7. The exterior angles of any polygon add to

   ANSWER $360°$

07 Homework

Homework — 5 tasks

Test every tiling claim by adding the angles at one meeting point.

1. Find each interior angle of a regular pentagon, hexagon and octagon.
2. Show by arithmetic which of them tile the plane alone.
3. Find a combination of two different regular polygons that meet at $360°$.
4. Find the number of sides of a regular polygon whose interior angle is $156°$.
5. Photograph or draw one piece of Uzbek ornament and name the polygons in it.